Stroma World — Tumor Calibration Knobs v1.0

> **Purpose:** Turn each of the 8 baseline traits into one or more tunable parameters the EA can read and the modeler can anchor to data. **Convention:** All continuous knobs are normalized to [0, 1]. Categorical knobs use named levels. **"PDAC Default"** = where the knob should sit for a canonical late-stage PDAC tumor cell before the EA touches anything.

Knob Anatomy

Each knob entry specifies:

Field	Meaning
Knob ID	Short reference name for code and XML config
Range	Allowed values
PDAC Default	Starting value for canonical PDAC genotype
Direction of danger	Which end makes the tumor harder to treat
Anchor source	Cell line data, patient/tissue data, or literature estimate — and why
Calibration note	What to look for when setting this value

Trait 1 — Constitutive Paracrine Secretion

Two knobs. One per stromagenic signal, because they have different druggable handles and the EA needs to modulate them independently.

Knob 1a: tgfb_secretion_rate

Field	Value
Range	0 (silent) → 1 (maximum observed PDAC secretion)
PDAC Default	**0.8**
Direction of danger	→ 1. Higher secretion = faster stroma buildup, deeper immune suppression.
Anchor source	**Cell line data (preferred).** ELISA measurements of TGF-β1 in conditioned media from PDAC lines (PANC-1, MIA PaCa-2, BxPC-3, Capan-1) are well-published and reproducible. Patient serum TGF-β levels exist but reflect systemic load, not local secretion rate.
Calibration note	Set 1.0 = highest published conditioned-media TGF-β concentration among PDAC lines (typically Capan-1 or PANC-1). Set 0 = normal ductal epithelial cells (HPDE). Default at 0.8 rather than 1.0 because not all PDAC cells are maximal secretors; leaves headroom for the EA to explore super-physiological secretion if needed.

Knob 1b: shh_secretion_rate

Field	Value
Range	0 (silent) → 1 (maximum observed PDAC secretion)
PDAC Default	**0.7**

Field	Value
Direction of danger	Ambiguous. Higher SHH = more stroma, but stroma also restrains tumor. The vismodegib paradox lives here.
Anchor source	**Cell line data (preferred).** SHH transcript levels (qPCR) and protein (Western/ELISA) from PDAC lines. PANC-1 and L3.6pl are high secretors; BxPC-3 is lower. Patient IHC staining intensity is available but harder to convert to a secretion rate.
Calibration note	Default slightly lower than TGF-β because SHH is the secondary stromagenic arm. The EA should discover that setting this to 0 (mimicking vismodegib) worsens outcomes without concurrent cytotoxic therapy. If it doesn't, the stroma-restraint feedback is broken.

Trait 2 — TGF-β Insensitivity

One knob. This is a switch, not a dial — but we allow intermediate values to represent heterogeneous SMAD4 status.

Knob 2: `tgfb_brake_sensitivity`

Field	Value
Range	0 (completely deaf — SMAD4 fully lost) → 1 (fully sensitive — SMAD4 intact, TGF-β arrests growth)
PDAC Default	**0.05**
Direction of danger	→ 0. Lower sensitivity = TGF-β builds stroma without braking tumor.

Field	Value
Anchor source	**Patient data (preferred).** SMAD4 IHC status from surgical resections is binary (intact/lost) and available in large cohorts (TCGA-PAAD, institutional TMAs). ~55% of PDAC is SMAD4-null. Cell lines are less informative because SMAD4 status is fixed per line and doesn't capture the partial-loss biology.
Calibration note	Default at 0.05 rather than 0.0 because even SMAD4-null cells retain trace SMAD-independent growth-inhibitory TGF-β responses. This is biologically minor but numerically prevents divide-by-zero edge cases in the brake calculation. Setting this to 0.5+ should flip the simulation into "TGF-β as tumor suppressor" mode — useful as a sanity check for the SMAD4 asymmetry.

Important: This knob controls *only* the growth-arrest arm of TGF-β signaling in tumor cells. The invasion/EMT arm (SMAD-independent) is governed by Knob 5a. The stromal response to TGF-β is not affected by this knob at all. These three pathways must remain decoupled.

Trait 3 — High Proliferative Rate With Broken Checkpoints

Two knobs. One for raw cycling speed, one for checkpoint responsiveness.

Knob 3a: `proliferation_rate`

Field	Value
Range	0 (quiescent) → 1 (fastest observed PDAC doubling time)
PDAC Default	**0.75**

Field	Value
Direction of danger	→ 1. Faster growth = more mass, more hypoxia, more paracrine pressure.
Anchor source	**Cell line data (preferred).** Doubling times for PDAC lines are extensively published: MIA PaCa-2 ~40h, PANC-1 ~52h, AsPC-1 ~60h, Capan-1 ~80h. These bracket the plausible range. Patient-derived organoid growth rates are emerging but less standardized.
Calibration note	Set 1.0 = fastest published PDAC line doubling time (~40h, MIA PaCa-2). Set 0 = no division. Default at 0.75 reflects that in vivo proliferation is slower than in vitro (nutrient/oxygen limitations), but still aggressive relative to stroma. The *ratio* of tumor proliferation to stromal cell proliferation matters more than the absolute value — aim for tumor cycling ≥3× faster than CAFs.

Knob 3b: checkpoint_integrity

Field	Value
Range	0 (all checkpoints destroyed) → 1 (fully intact checkpoints, normal cell)
PDAC Default	**0.05**
Direction of danger	→ 0. Fewer checkpoints = unstoppable growth, resistant to contact inhibition and DNA damage arrest.
Anchor source	**Patient data (preferred).** CDKN2A deletion frequency (~90%), TP53 mutation frequency (~75%), and RB1 pathway status are well-characterized in TCGA-PAAD and ICGC. The combinatorial checkpoint landscape is better captured by patient genomics than by any single cell line.

Field	Value
Calibration note	This is an *aggregate* knob representing the net state of CDKN2A + TP53 + RB1. It doesn't replace the Boolean network — it's the post-combination-rule output from the Gene State Representation (Part 4, Veto Rule 1). When CDKN2A=LOF AND TP53=LOF AND MYC=GOF, the combination rules force this to NONE, which maps to 0.05 here. The 0.05 floor prevents cells from being literally immortal in the simulation (some stochastic death must remain).

Trait 4 — Apoptotic Resistance at Baseline

One knob. Represents the net balance of pro- and anti-apoptotic machinery.

Knob 4: `apoptosis_resistance`

Field	Value
Range	0 (maximally sensitive — normal apoptotic response) → 1 (maximally resistant — apoptosis essentially disabled)
PDAC Default	**0.85**
Direction of danger	→ 1. Higher resistance = tumor persists longer, sustains paracrine secretion, survives partial drug exposure.
Anchor source	**Cell line data (preferred).** IC50 curves for gemcitabine, TRAIL, and ABT-263 (navitoclax) across PDAC lines directly measure apoptotic threshold. BxPC-3 is relatively sensitive; PANC-1, MIA PaCa-2, and AsPC-1 are resistant. The ratio of BCL-XL to BAX protein levels (Western blot) is published for major lines.

Field	Value
Calibration note	This knob is the phenotype output of the BCL-XL vs. BAX competition. When BCL-XL=GOF and BAX=WT, the combination rules produce HIGH death resistance, mapped here to 0.85. Full 1.0 is reserved for cells that have *also* activated NRF2 and ABCB1 under stress (Knob 7). Baseline resistance should be high enough that spontaneous apoptosis is rare, but low enough that a sufficiently concentrated drug *can* still kill.

Trait 5 — Inducible EMT

Two knobs. One for the threshold to trigger EMT, one for the phenotypic extent once triggered.

Knob 5a: emt_induction_threshold

Field	Value
Range	0 (triggers on any trace signal) → 1 (requires overwhelming signal to trigger)
PDAC Default	**0.3**
Direction of danger	→ 0. Lower threshold = EMT happens easily, more invasive front, more PD-L1.
Anchor source	**Cell line data (preferred).** TGF-β dose-response curves measuring E-cadherin loss or vimentin gain in PDAC lines. PANC-1 undergoes EMT at ~5 ng/mL TGF-β; BxPC-3 is more resistant. Dose-response data directly maps to a threshold value.
Calibration note	This controls the *signal concentration* (TGF-β or hypoxia severity) required to flip ZEB1=ON. At 0.3, EMT is relatively easy to induce — consistent with PDAC's known mesenchymal tendency. Cells in the oxygenated, low-TGF-β tumor core should remain below threshold; cells at the

Field	Value
	hypoxic, stroma-adjacent periphery should exceed it. If the entire tumor goes mesenchymal at once, the threshold is too low. If no cells undergo EMT, it's too high. Spatial heterogeneity of EMT status is the validation criterion.

Knob 5b: emt_phenotype_extent

Field	Value
Range	LOW / MEDIUM / HIGH
PDAC Default	**HIGH**
Direction of danger	→ HIGH. Greater extent = more motility gain, more adhesion loss, more MMP2, more PD-L1.
Anchor source	**Cell line data (preferred).** Transwell migration assays and Matrigel invasion assays post-TGF-β treatment. MIA PaCa-2 and PANC-1 show dramatic motility increases post-EMT. Wound-healing assay closure rates are a secondary source.
Calibration note	This is categorical, not continuous, because EMT in PhysiCell maps to discrete phenotype parameter sets (motility speed, adhesion strength, secretion profile). LOW = partial EMT (some motility gain, E-cadherin partially retained). MEDIUM = classical EMT (full E-cadherin loss, moderate motility). HIGH = complete mesenchymal shift (maximum motility, minimum adhesion, MMP2 active, PD-L1 upregulated). PDAC defaults to HIGH because the KRAS/TP53/SMAD4-null background is among the most EMT-permissive genotypes known.

Trait 6 — Hypoxia-Responsive Phenotype Switching

Two knobs. One for the oxygen threshold, one for the magnitude of the phenotype shift.

Knob 6a: hypoxia_response_threshold

Field	Value
Range	0 (HIF1A activates at any O_2 dip) → 1 (requires near-anoxia to activate)
PDAC Default	**0.35**
Direction of danger	→ 0. Lower threshold = larger hypoxic zone, more HIF1A-driven behaviors across the tumor.
Anchor source	**Literature estimate (neither cell line nor patient data alone is sufficient).** HIF1A stabilization occurs at ~1–2% O_2 in most cell types, but PDAC cells show HIF1A activity at higher O_2 levels (~3–5%) due to KRAS-driven HIF1A transcriptional priming. The threshold is a biophysical parameter modified by genetic context. Published O_2 measurements in PDAC xenograft tumors (pimonidazole staining, Eppendorf electrode) provide the tissue-level reference.
Calibration note	Map 0.0 = HIF1A ON at 10% O_2 (essentially always active). Map 1.0 = HIF1A ON only below 0.5% O_2 (near-anoxic). PDAC default at 0.35 corresponds to ~2–3% O_2 threshold, reflecting KRAS-primed HIF1A sensitivity. This knob determines how much of the tumor volume is "hypoxic" at any given time, which in turn determines the size of the EMT-active, stress-adapted, stroma-amplifying zone.

Knob 6b: `hypoxia_phenotype_shift`

Field	Value
Range	LOW / MEDIUM / HIGH
PDAC Default	**HIGH**
Direction of danger	→ HIGH. Greater shift = more aggressive adaptation to hypoxia, stronger positive feedback loop.
Anchor source	**Cell line data (preferred).** Hypoxia chamber experiments (1% O_2, 24–72h) measuring fold-change in motility, secretion rates, and drug resistance in PDAC lines. PANC-1 and AsPC-1 show dramatic phenotype shifts; Capan-2 is more modest.
Calibration note	LOW = mild motility increase, slight TGF-β boost. MEDIUM = moderate motility increase, measurable drug resistance gain, ZEB1 activation likely. HIGH = full phenotype switch — motility surge, secretion amplification, NRF2 priming, EMT induction if threshold is met. PDAC defaults to HIGH because the KRAS-primed HIF1A context makes PDAC cells exceptionally responsive to hypoxia compared to other carcinomas.

Trait 7 — Drug-Inducible Efflux

Two knobs. One for the delay before efflux activates, one for the strength of efflux once active.

Knob 7a: efflux_induction_delay

Field	Value
Range	0 (instant efflux — constitutive resistance) → 1 (very slow induction — long vulnerability window)
PDAC Default	**0.5**
Direction of danger	→ 0. Faster induction = shorter window for drug to kill before resistance kicks in.
Anchor source	**Cell line data (preferred).** Time-course experiments measuring ABCB1 protein or mRNA induction after gemcitabine exposure in PDAC lines. Published data typically shows detectable upregulation at 12–24h, maximal expression at 48–72h. NRF2 nuclear translocation kinetics provide the upstream timing reference.
Calibration note	This knob determines the *time gap* between drug arrival and efflux activation. At 0.5, there is a meaningful window where the drug can act before ABCB1 ramps up — consistent with the clinical observation that first-dose chemotherapy has the strongest effect and subsequent doses face increasing resistance. The EA should discover time-pulsing strategies that exploit this window. If it doesn't, the delay may be set too short.

Knob 7b: efflux_strength

Field	Value
Range	0 (efflux has no effect even when active) → 1 (efflux completely neutralizes drug)

Field	Value
PDAC Default	**0.7**
Direction of danger	→ 1. Stronger efflux = drug is more effectively expelled, narrower kill window.
Anchor source	**Cell line data (preferred).** Rhodamine-123 efflux assays or calcein-AM assays measuring ABCB1 functional activity in resistant vs. parental PDAC lines. Fold-change in IC50 between ABCB1-high and ABCB1-low sublines directly calibrates this knob.
Calibration note	At 0.7, efflux substantially reduces intracellular drug concentration but doesn't eliminate it entirely — high enough drug delivery can still overcome resistance. This forces the EA to find combination strategies (stroma breaching + high-dose delivery + efflux inhibition) rather than relying on any single approach. Setting this to 1.0 would make drug-adapted cells completely unkillable, which is unrealistic — even MDR1-high cells can be killed at sufficient concentrations.

Trait 8 — Density-Driven ECM Compaction

One knob. This is a mechanical parameter, not a genetic one.

Knob 8: `mechanical_compaction_strength`

Field	Value
Range	0 (tumor growth exerts no mechanical effect on surrounding ECM) → 1 (maximum observed solid stress in PDAC)

Field	Value
PDAC Default	**0.6**
Direction of danger	→ 1. Stronger compaction = higher interstitial pressure, more drug exclusion, more vascular collapse.
Anchor source	**Patient data (preferred).** Solid stress measurements from freshly resected PDAC specimens (Stylianopoulos, Jain et al.) using tumor-opening assays and compression testing. PDAC has among the highest solid stress of any tumor type (~6–10 kPa). Cell line data is not informative here — this is a tissue-level emergent property that doesn't manifest in monolayer culture. Xenograft mechanical measurements are an acceptable secondary source.
Calibration note	This is the only knob in the set that is purely mechanical. It modifies how tumor cell proliferation-driven volume expansion translates into local ECM density increase at the tumor-stroma boundary. At 0.6, compaction is significant but not the dominant barrier-building mechanism (CAF-mediated ECM production is still stronger). At 1.0, compaction alone would create a dense boundary even without CAF activation. The real PDAC situation is a combination of both, with CAF activity dominant.

Master Knob Table

ID	Knob	Range	PDAC Default	Danger→	Anchor
1a	tgfb_secretion_rate	0–1	0.8	→1	Cell line
1b	shh_secretion_rate	0–1	0.7	Ambiguous	Cell line
2	tgfb_brake_sensitivity	0–1	0.05	→0	Patient

ID	Knob	Range	PDAC Default	Danger→	Anchor
3a	proliferation_rate	0–1	0.75	→1	Cell line
3b	checkpoint_integrity	0–1	0.05	→0	Patient
4	apoptosis_resistance	0–1	0.85	→1	Cell line
5a	emt_induction_threshold	0–1	0.3	→0	Cell line
5b	emt_phenotype_extent	L/M/H	HIGH	→H	Cell line
6a	hypoxia_response_threshold	0–1	0.35	→0	Literature
6b	hypoxia_phenotype_shift	L/M/H	HIGH	→H	Cell line
7a	efflux_induction_delay	0–1	0.5	→0	Cell line
7b	efflux_strength	0–1	0.7	→1	Cell line
8	mechanical_compaction_strength	0–1	0.6	→1	Patient

Totals: 13 knobs. 10 continuous (0–1). 2 categorical (L/M/H). 1 ambiguous-direction (SHH).

Anchor breakdown: 9 cell line, 3 patient, 1 literature estimate.

Anchoring Strategy

Cell-line-anchored knobs (9 of 13)

These measure single-cell or monolayer behaviors that are directly observable in vitro: secretion rates, proliferation times, IC50 values, migration speeds, efflux kinetics. The mapping from measurement to [0,1] is straightforward.

Priority lines for calibration (rank-ordered by relevance):

Line	KRAS	TP53	SMAD4	CDKN2A	Best represents
PANC-1	G12D	R273H	WT	HomDel	Moderate PDAC, SMAD4-intact variant
MIA PaCa-2	G12C	R248W	WT	HomDel	Aggressive, fast-growing PDAC
AsPC-1	G12D	null	null	HomDel	Late-stage, fully checkpoint-deficient
BxPC-3	**WT**	Y220C	HomDel	HomDel	KRAS-wild-type outlier — useful as control
Capan-1	G12V	A159V	WT	HomDel	Well-differentiated, high TGF-β secretor

Recommendation: Calibrate against AsPC-1 as the primary reference (closest to canonical late-stage genotype: KRAS+, TP53−, SMAD4−, CDKN2A−). Use PANC-1 as the SMAD4-intact comparison for validating Knob 2.

Patient-data-anchored knobs (3 of 13)

These measure population-level or tissue-level properties that don't manifest in vitro: mutation frequencies, checkpoint loss combinations, solid stress.

Knob	Data source	Key dataset
tgfb_brake_sensitivity	SMAD4 IHC status	TCGA-PAAD (n=185), institutional TMAs
checkpoint_integrity	Co-occurrence of CDKN2A/TP53/RB1 alterations	TCGA-PAAD, ICGC-PACA-AU
mechanical_compaction_strength	Solid stress measurements	Nia et al. 2017 (*Nat Biomed Eng*), Stylianopoulos et al. 2012 (*PNAS*)

Literature-estimated knobs (1 of 13)

Knob	Why not cell line or patient	Best available reference
hypoxia_response_threshold	O_2 threshold for HIF1A is a biophysical parameter modified by genetic context; neither cell line (measured at atmospheric O_2) nor patient data (no single-cell O_2 resolution) captures it cleanly	Koong et al. 2000 (pimonidazole in PDAC), Erkan et al. 2012 (hypoxia review), McKeown 2014 (O_2 thresholds review)

What the EA Is Allowed to Touch

Not all 13 knobs are in the EA's search space. Some are **fixed background** (tumor identity) and some are **intervention targets** (what the EA optimizes).

Category	Knobs	Rationale
Fixed (tumor identity — EA cannot change)	2, 3a, 3b, 4, 8	These define what PDAC *is*. The EA doesn't get to make the tumor slower-growing or more apoptosis-sensitive — it has to deal with the tumor as it exists.
Observable (EA can read, not write)	5a, 5b, 6a, 6b	EMT and hypoxia response are emergent from the microenvironment. The EA can *indirectly* affect these by changing stroma density or drug timing, but can't directly set the thresholds.
Targetable (EA search space)	1a, 1b, 7a, 7b	These are the druggable handles. The EA can test interventions that reduce TGF-β secretion, block SHH, shorten the efflux delay (efflux inhibitors), or weaken efflux strength. The EA's job is to find the combination of interventions on these knobs that maximizes tumor kill.

This partition prevents the EA from "cheating" by evolving biologically impossible tumors. It can only manipulate what a drug or intervention could plausibly change.

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13 knobs. 0–1 scales. Every one traceable to a specific data source. The EA gets 4 of them. The tumor keeps the rest.